Raising the Floor or Changing the Test?

AI-assisted course design and student outcomes in two community college mathematics courses

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Under review, *Teaching Mathematics and its Applications*

The question instructors ask first

Instructors now use AI to build course materials. Studies describe the materials, or worry about student cheating. Almost none ask what happens to the students.

Three plain questions:

1. After I redesigned each course with AI, did students score higher on the proctored final, and did as many take it?
2. Which students gained: the weakest, the strongest, or everyone?
3. Did homework and quiz scores still tell students who was ready for the final?

An average can hide what matters most at an open-access college: which students moved.

Four classes, one instructor

Course	Term	Took the final	Enrolled
Introductory statistics	Winter 2025, before AI	28	29
	Winter 2026, with AI	27	27
College algebra (STEM)	Spring 2025, before AI	32	34
	Spring 2026, with AI	19	26

- A large urban community college in Southern California; every section had a corequisite support section.
- Gradebook records only: no names, no demographics, no placement or repeat status.
- I taught and graded every section. This is a practitioner case study, not an experiment.

AI did two different jobs

Statistics: harder exams

AI helped me write more demanding exams, lectures and study guides. Higher-order multiple-choice questions on the final rose from 18% to 48%. Quizzes still came from a question bank.

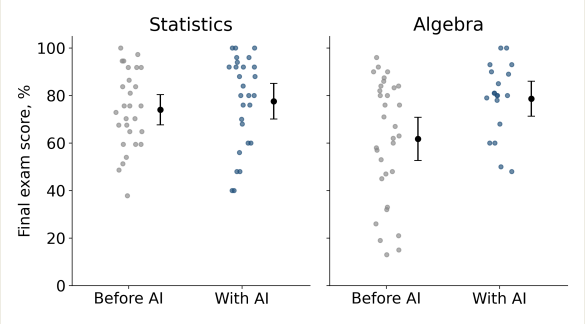
Algebra: one aligned course

AI helped me build quizzes to match the exams, step prompts under each problem, a formula sheet, and a practice final.

In both, I revised examples and scaffolds during the term for the students in front of me. I reviewed and approved every item.

AI-assisted design is not one intervention.

Algebra scores rose; statistics held on a harder exam



61.8% algebra before

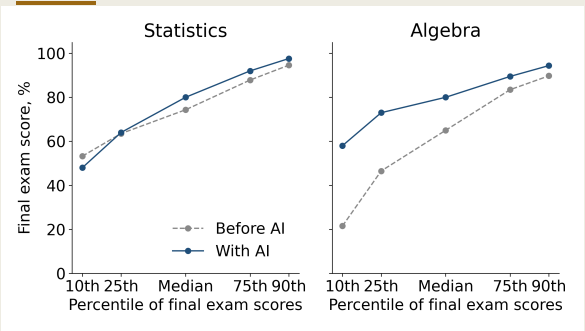
78.7% algebra with AI

Average final exam score among students who took it (95% CI for the gain 5.5 to 28.3 points).

Statistics: 74.0% before, 77.6% with AI, on a much more demanding exam.

The algebra final also changed from free response to mostly multiple choice (slide 7).

The gain was largest at the bottom



In algebra:

- 10th percentile: about 22 to 58.
- 90th percentile: 90 to 94, barely moved.
- Below 50%: 10 of 32 students before, 1 of 19 with AI.

As scored, the students most likely to fail moved the most.

But the test also changed

The earlier algebra final was 20 free-response problems. The AI final was 19 four-choice problems and one proof.

On multiple choice, guessing alone earns points, and it lifts the lowest scores most. Removing the average blind-guessing score (23.8 points):

Algebra change	As scored	Guessing points removed
10th percentile	+36 points	+23 points
Average	+16.9 points	+10.3 points
Share below 50% with AI	5%	21%

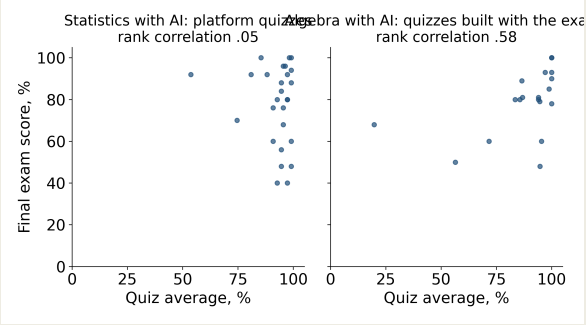
The pattern stays largest at the bottom, but the gain is no longer clear.

Who took the final

- Statistics: nearly everyone took the final in both terms.
- Algebra: 94.1% of enrolled students took it before, 73.1% with AI.
- Pass rate among all enrolled algebra students (70% or higher): 18 of 34 before, 15 of 26 with AI. About the same.
- Worst case, that the students who skipped were exactly the weakest: the gain shrinks to 6.0 points (95% CI -3.5 to 15.4).

Count who finishes, not only who scores.

Practice scores crowded toward 100%



Homework at 95% or higher: 32% to 85% in statistics, 22% to 63% in algebra.

Quiz and final exam rank correlation with AI:

- Question-bank quizzes (statistics): .05
- Quizzes built with the exams (algebra): .58

The courses already differed before AI, algebra was my second AI term, and the comparison is too small to settle.

What this means for practice

- **Look at the bottom of the distribution**, and hold the exam format fixed when you compare.
- **Build quizzes together with the exams.** Practice left outside the design can stop measuring readiness.
- **Watch two numbers every term:** the share of scores at the ceiling, and the quiz and exam correlation. Both come from any gradebook.
- **Count who finishes,** not only who scores.

The hope, not a promise

If AI takes over part of the routine work of building materials, instructors may have more time for every student. At an open-access college, the hope is that this time reaches the students who need it most.

The same tools can also widen gaps in access, carry bias, and invite students to lean on them in place of learning.

These results fit the hope in direction but cannot confirm it.

The hope is worth testing directly: same exam format, the instructor's time measured, results for the students most at risk.

What this study cannot show

- **One instructor, intact classes.** No random assignment; my own growth as a teacher is part of the change.
- **Different exams,** including the change from free response to multiple choice.
- **Small classes:** 19 to 32 students per class, so intervals are wide.
- **Student AI use was not measured.**

These are associations from one instructor's classes, not proof that AI caused them.

The takeaway

AI-assisted design was followed by fewer very low scores among the students who took the algebra final. This study cannot say how much came from the design and how much from the new exam format.

The next test should hold the exam format fixed, build the whole course to fit together, and read results across the distribution, not only on average.

Thank you. Safaa Dabagh learnwithoutwalls.com
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Earlier version: EdArXiv, doi.org/10.35542/osf.io/x9nkw_v1